SPE-229516-MS

Accelerating Ensemble Based Field Characterization Enabled by Grid-Less Multi-Resolution Model

D. Das, S. Marzouk, and R. Kedia, Halliburton

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

The Project objective is to address an inherent challenge of grid-based model where at the first place the modeler needs to lock the resolution with the grid scale. Due to locking the grid resolution, the log and seismic data needs to be upscaled in that resolution, the original resolution and information gets lost in the model.

Grid-less Scalable Earth Modeling is an innovative agile modeling approach that uses all the available data to produce property models that are consistent across all resolutions and all scales. It is a point-based modeling technology; the need to build grids and populate them with properties is deferred until flow simulation or other grid-based processes are run.

This grid-less approach has been attempted to build a multi-resolution model which can host multi-resolution model in a shared earth container. In order to demonstrate the power of the grid-less modeling, an extended area has been taken into consideration beyond the field area as the project area of interest. This AOI includes the main field of target with a significant no. of wells (approximately 60 wells) along with a potential near field exploration area towards the south and extended towards North for any future exploration process. The objective of selecting such a mix of areas with variable data cluster is to showcase the value in terms of the below key points:

A. This modeling approach includes a region based VPC modeling to replicate the geological concept in a 3D space away from the well cluster to minimize the biasness of the clustered wells
B. Scalability of grid-less modeling to exhibit the value of downscaling and zooming in the model as and when required to incorporate any geological details and for multi-purpose approach like Field Development, Near Field Exploration and new exploration venture

Introduction

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The second objective is to break the silo from Geoscience to Reservoir Engineering domain and build an Ensemble based workflow which can cater for the uncertainty across different domains and derive 80-100 Ensemble models through a data driven approach. The static model built in the aforementioned grid-less approach has been transferred to this Ensemble Modeling workflow orchestrator capturing the uncertainty from various disciplines ranging from geophysics, Petrophysics, Geological Modeling, Reservoir Engineering etc. to come up with a pair of history matched static and dynamic models. The key objective of the introduction of this workflow are as below:

A. Introducing the value of bridging the gap between the domains like G&G to RE in the uncertainty space for more informed decision in a probabilistic way
B. Accelerating the decision process through workflow orchestration and establishing a disruptive approach to have the consistency between the static and dynamic model substantiated by the observed data

In the below section, the workflow adopted to achieve these above objectives are explained in detail.

Methodology

The methodology for this Ensemble based modeling via grid-less approach has been sub-divided into 2 major workflow segments as follows:

- Grid-less Static Modeling (Scalable Earth Modeling)
- Unified Ensemble Modeling (Integrated Uncertainty Analysis)

Scalable Earth Modeling

This is the grid-less modeling approach which has been adopted for this Area of Interest to demonstrate the value of not having grids at geological modeling stage. This workflow generated various scenarios with an evergreening dynamic scaling approach to capture the insights of the reservoir.

The methodology includes the below key steps to combine the information for Petrophysical evaluation to the seismic attribute analysis in a geostatistical model.

- ***Structural Framework:*** A dynamic framework has been constructed using the seismic interpretation (horizons and faults) to create the target reservoir zone calibrated with the well picks followed by fault block creation with fault compartmentalization. Approximately 54 faults have been incorporated in the framework building process to create the major fault blocks in the central field area. One of the key benefits of this framework is to integrate any new data as and when available to update the framework on-the-fly.
- ***Seismic Attribute Analysis:*** Detail Seismic Attribute Analysis had been performed to extract the insights about the reservoir continuity. These results are important in 2 ways to build an informed geological model.
- Building a better horizontal variogram in the parts of the model where the well density is less
- Support the Anisotropy direction identification from seismic trend
- Also, the seismic attributes can be used as a probability trend to guide the population of the facies properties based on the geostatistical parameters

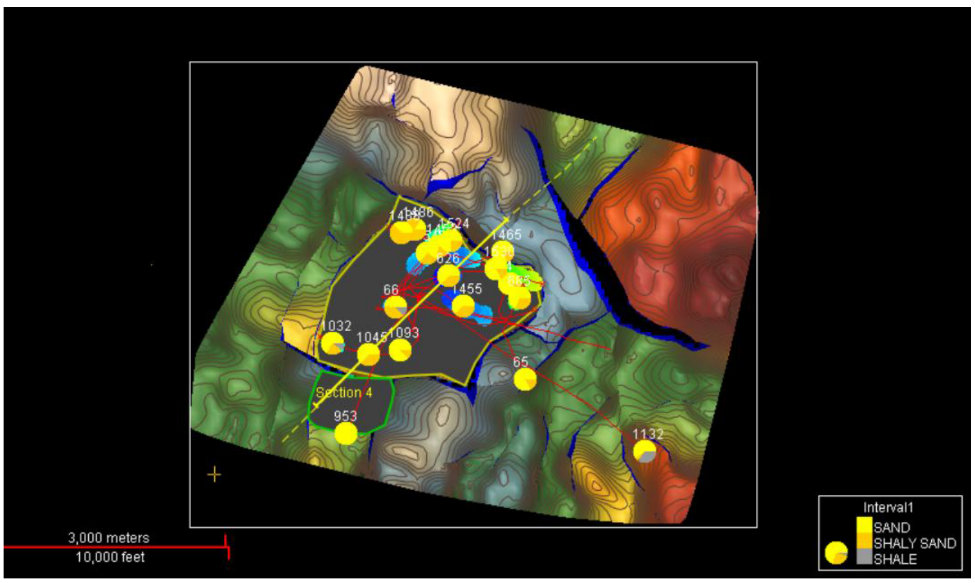

Figure 1—The structured Depth Map (with fault polygons) of the top reservoir being displayed with the key wells and the respective facies fraction from the reservoir interval

Various seismic attributes have been run through parameterization to establish a better correlation between the facies and seismic attributes. The key seismic attributes that have been generated are RMS amplitude, Sweetness, Instantaneous Frequency, Reflection Strength and Relative Acoustic Impedance etc. In order to extract the maximum insights from seismic attributes, a multi-variate analysis has been run utilizing all the most relevant attributes which can contribute to the reservoir continuity trend prediction. A spectral decomposition analysis also has been conducted to extract the dominant frequencies which are denoting the key geological features.

- ***Facies Model Building:*** The key differentiator of this workflow comes into the picture at this stage while we skipped building a full-scale grid with a specific locked resolution and straight away moved to loading the well data and conditioning seismic data in the modeling space (UVW) from their original XYZ space. The variogram modeling workflow was performed in this space with the original resolution of the data (0.152 m) of the well for facies log interpretation. In this model we used 3 major facies (Sand, Shaly Sand and Shale) based on the interpretation given from the petrophysical evaluation. The major, minor and vertical range have been derived from the well and background seismic along with the anisotropy direction of around -59° which is well aligned with the background depositional trend. Also, an innovative modeling approach using "Region Based VPC" has been used in this model to establish the geological concept better especially in the area with very limited well control. The production data from the field clearly justifies the presence of shale barrier towards the northern side of the area of interest which needs to be replicated in the model with the geological concept. Hence, we prepare a set of polygons to define the entire area based on the geological depositional trend supported by the background seismic and the well & production data. Those polygons were used to cluster representative wells into individual Vertical proportion Curve (VPC) for each polygon. Following that, we replicated the data beyond the polygon to control the data spread in the area with limited well control and introduced that barrier shale at the top layers as substantiated by the production data. These VPCs were then further combined into a Grouped Proportion Curve model through Tau gridding algorithm which created Geological trend model in the background. This model was used in the Facies algorithm where we used "Truncated Gaussian Simulation" (TGS) to populate the facies properties with standard geological rules of facies occurrence. Various scenario models were tested with the background Grouped Proportion Curve Trend model and Seismic attributes to finalizing the best model to proceed for the next step of Petrophysical modeling.

- ***Petrophysical Property Modeling (Porosity and Permeability):*** We build a full-scale Geostatistical model of Porosity and Permeability using a "Turning Band Simulation" (TBS) which can be parallelized in the processors to build high-resolution large models in relatively shorter time. Additionally, variogram model has been used for this petrophysical property distribution constrained with facies model and the histogram fraction which is originally extracted from the high-resolution log data not from an upscaled and averaged information unlike grid-based model.
- ***Saturation Modeling:*** Saturation Height Function has been generated for the 2 key major reservoir facies (Sand and Shaly Sand) to be included in this grid-less modeling approach. The Saturation Height Function also was created in an attempt to capture the potential transition zone between the Oil and Water zone at the OWC. The Saturation Heigh Function was directly applied on the static model to create a Saturation model from where the logs were extracted to have QC with the function which was earlier calibrated with MICP data.
- ***Net-To-Gross Model:*** A Net Pay equation given by the Petrophysicist was applied on the full static model integrating the Porosity, Permeability and Saturation to create the Net Pay 3D model out of it.
- ***Scalable Visualization creation:*** Here the biggest differentiator of this modeling approach come which provides the flexibility to the modeler to generate multiple multi-purpose and multi-resolution visualization of model from the same "Grid-less model" without any requirement of building multiple models in siloed and repetitive manner. The detail of the multi-resolution model is discussed in the results and observations section of the manuscript.
- ***Grid-less Volumetrics:*** Finally, as the last but not least step of the static modeling workflow, the grid-less modeling approach was performed to test various scenarios of Development field or Near Field Exploration at the south or the Northern exploration area alongside different investigative scenarios of multi- resolution.
- ***Grid creation:*** In order to run the simulation the final Scalable Earth Model has been transferred into a grid with resolution of 50mX50M with vertical resolution of 2m with proper averaging methods respective to individual parameters such as "Most of" method for Facie, "Arithmetic average" for porosity, "Harmonic average" for Permeability etc.

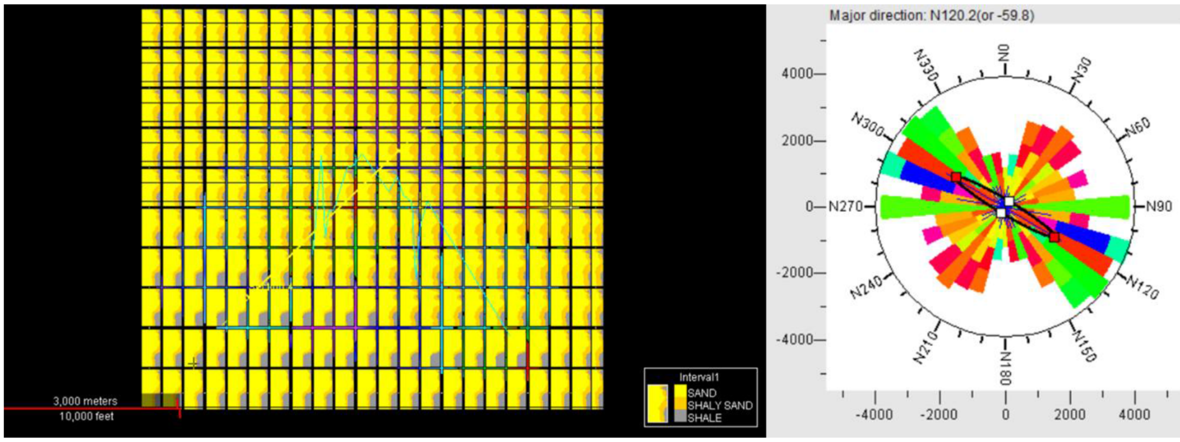

Figure 2—The GDE trend model result from the "Region Based Vertical Proportion Curve" highlighting the presence of shale towards Northern side substantiated by the production data and the Horizontal variogram map extracting the anisotropy direction from the data (-59.8)

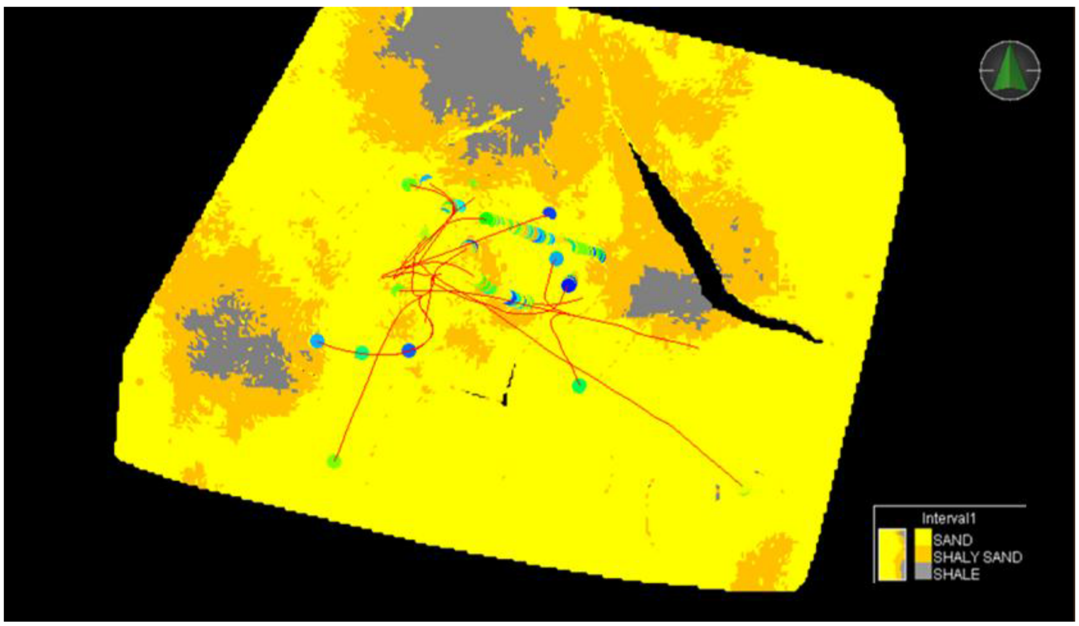

Figure 3—The grid-less model visualization for the entire AOI including the central and south field and the northern exploration area with a resolution of 25m × 25m × 1m

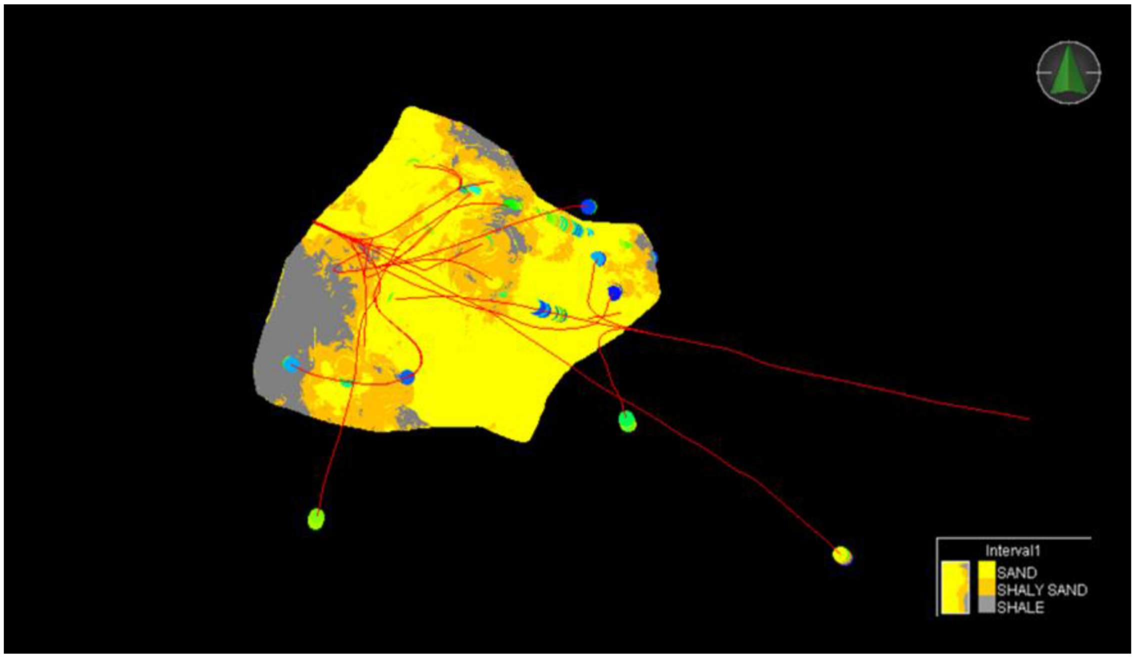

Figure 4—The grid-less model visualization for the central field area with a resolution of 10m × 10m × 1m (showing the power of zoom-in functionality)

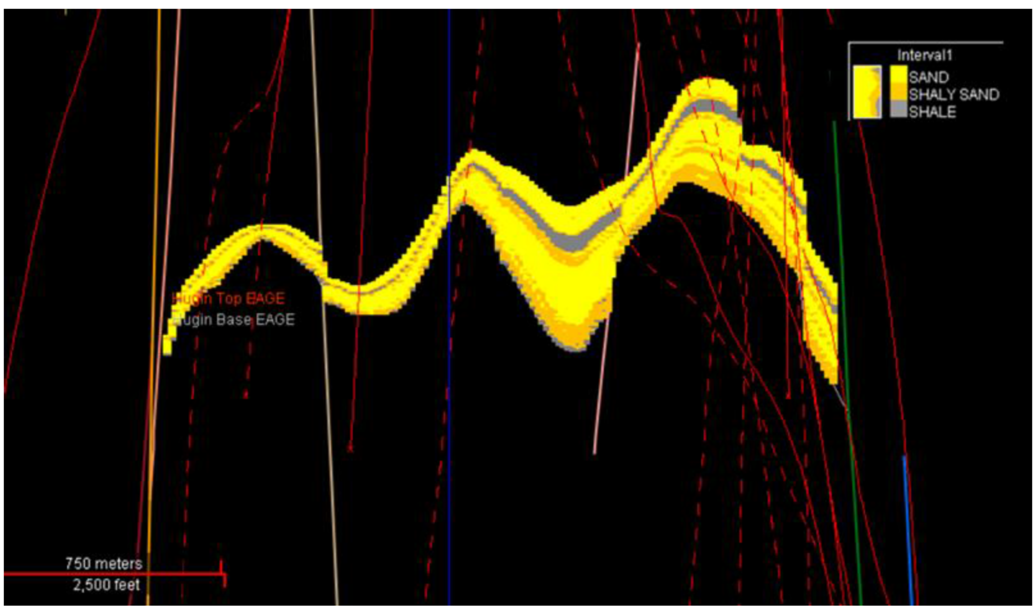

Figure 5—A cross section showing the shale distribution in the central field area constrained by the background GDE model (Derived from the region based VPC)

Unified Ensemble Modeling. Following that the ensemble-based model has been run integrating the uncertainty ranges from geophysical interpretation, modeling parameters, petrophysical properties etc. in combination with the reservoir uncertainties pertaining to relative permeability, saturation etc. to generate ensembles calibrated with observed data for the history matching and subsequent production forecasting.

A fundamental change in thinking is needed in reservoir development, putting uncertainty at the center of the modelling exercise and risk at the center of the decision-making process. Unified Ensemble Modeling (UEM) is the approach developed by Halliburton to make that change happen.

At any given time during the lifecycle of an asset, the perception of the reservoir is uncertain and will change as new data is acquired and we continue to learn. In this 'uncertainty-centric' approach we can determine the likelihood of reservoir parameters (e.g. porosity, permeability) having a certain value using the observed data as a constraint. As such ensemble-based modeling is solved as an inverse rather than a forward problem.

- ***Uncertainty Parameterization (Static):*** Static model Uncertainty parameterization involves ranges of uncertainty associated with the Structural interpretation, Petrophysical log interpretation (like Porosity, permeability etc.), Static modeling parameters (ex. Variogram, probability maps/ cubes, anisotropy direction etc.)
- ***Uncertainty Parameterization (Dynamic):*** The Dynamic model uncertainty ranges include the uncertainty associated with the properties like Relative Permeability, Saturation Height Function, Capillary Pressure, Aquifer etc.
- ***Prior Ensemble Creation:*** The Prior Ensemble models are the results of orchestrated simulation of static and dynamic modeling together taking all the uncertainties into account. At this stage we don't include any observed data to constrain the results. The whole objective of building this prior ensemble modeling is to verify the uncertainty range of the input parameters in terms of retaining the consistency of the range and trend.
- ***Posterior Ensemble Modeling:*** At the posterior ensemble modeling stage, the observed data were incorporated to train models through a multi-data assimilation (MDA) process with Ensemble smoother and capturing the geological consistency through Graph Neural network algorithm. This posterior ensemble models take the observed data like production data, water cut etc. into account to build the simulated results and automate the history matching process without manual adjustment or local optimization of hands full of parameters.
- ***Integrated Reservoir Management Analytics:*** Once around 80 Ensemble models are created which are equiprobable in nature, in order to evaluate the Ensemble models to make any decision, various analytical assessments have been performed. Constraining the results with multiple filters and combining the Ensemble models in a statistical way, the connected volumes, sweet spot maps were created which have been contributed from all the 80 ensemble models instead of a single most likely model.

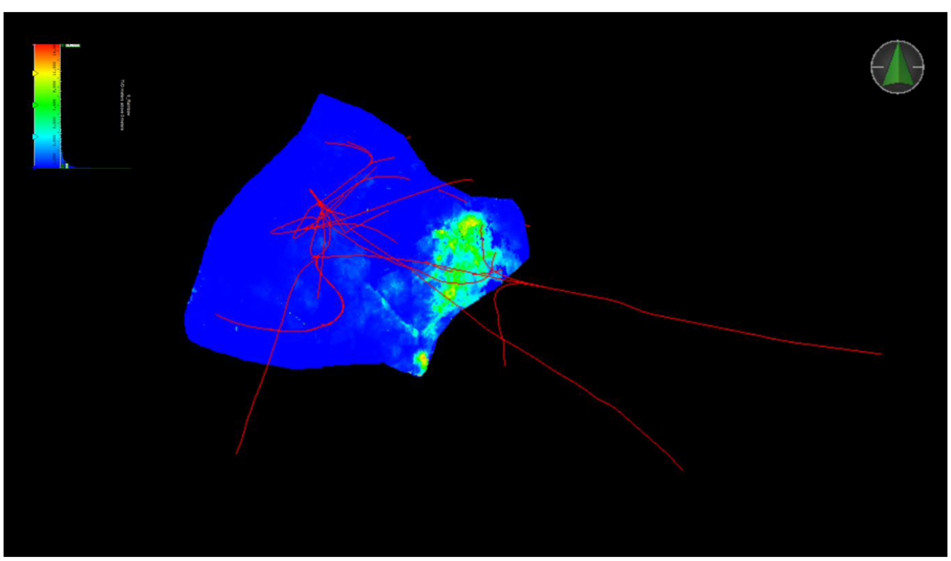

Figure 6—Ensemble driven Calibrated Permeability model from 80 Ensembles restored in a grid-less model to complete the feedback loop

Performance Benchmarking

The performance of the grid-less model has been compared with the grid-based model. There are 2 ways it brings the value on the table:

1. ***High-resolution Faster model:*** The high-resolution model at the central field area has around 8M points (with a resolution of 10mX10mX1m) takes around 5 mins to run the facies, Porosity and Permeability property simulation versus the same scale of model is taking 30+ mins to run all the models. On top of it the high-resolution grid model cannot take into account the data outside the central field whereas the grid-less model always takes into account the all the data from the entire AOI to keep the model consistent with the entire area.
2. ***Resolution optimization:*** The resolution optimization has been done based on selecting the individual areas of interest like Central Field, South field visualization extraction at various resolution like 25mX25mX1M and 10mX10mX1m etc.

Results and Conclusion

The study area for this project has been taken into account to show that based on the requirement in the course of field life cycle if there is a requirement of changing the resolution at any interval is required, the same grid-less model can be utilized as an evergreen model without any requirement of building a new model in silo from scratch. A full-scale geostatistical model has been built guided by seismic data and getting control from a region based VPC model to depict the consistency from regional geology to field scale. The Ensemble based modeling approach generates 80-100 ensembles to predict the production forecast and connected volume for the movable hydrocarbon sweet spot. The ensemble models are equiprobable models which are combined together to calculate the probability of each value for each cell of the model in terms of static and dynamic properties.

The Static model generates a grid-less scalable model which can be viewed in any resolution and any extent as required by the geo-modeler. For the scenario purpose, we tested a full model visualization in the resolution of 100mX100mX4m which includes the entire field, near field exploration area (Southern part) and the rest Exploration area (Mainly the northern part) to evaluate the regional model can capture the transgressive shale towards the northern side of the model. Then for the potential volumetric assessment purpose in the newly discovered near field exploration area, a high-resolution model was extracted with the resolution of 50mX50mX2m. Finally, a full-scale model has been extracted from the central field area with the resolution of 10mX10mX0.5m to achieve the most accurate volumetrics and fault structural control. The grid-less volumetric was run for both the central and south area at their respective resolution.

The consistency of the individual extracted visualization of the 3 model areas have been evaluated which exhibits a consistency as while generating individual visualization for each model area, the background geo-statistics would always take all the available well and seismic in the calculation. That delivers the geological reasonable models with the full consistency across the fields.

The final grid model from the central field area has been provided to the Reservoir Engineer with all the possible uncertainty ranges to run the Unified Ensemble Model. The individual parameters were given different distribution of the ranges such as normal, log-normal (For Permeability) etc. Total 80 history matched Ensemble models were generated with all the various plots of Water cut, Production profile, Pressure plot etc. The hard observed data were used to calibrate the ensembles for the history matching. Unlike conventional approach of doing uncertainty analysis in static and dynamic domains separately, this approach of Unified Ensemble modeling generates pairs of history matched static and dynamic models which in turn bridges the gap between G&G team and Reservoir team to have a consistent model to pursue further decision-making process at field development stage.

Using the Integrated reservoir Management workflow, few filters were applied of the 80 ensemble models that were created in the last step. The key geological filters that were used in combination of "AND" to refine the Ensemble models are as below:

- Porosity: >10%
- Permeability: >1 darcy
- Water Saturation (Sw): <70%

These applied filters help to screen the ensemble model results which adhere to the filter brackets provided and run various statistical plots such as Mean scenario, Most Likely scenario on the screened-out Ensembles to prepare the below property models or maps:

- Connected Volume
- Sweet Spot Map
- Movable Oil Volume
- Automated Target generation

These maps and sub-surface models help to prepare a full field well planning with the identified targets in collaboration with the surface, sub-surface and shallow hazard analysis in a collaborative well planning platform. However, for this project the well planning was not part of the scope.

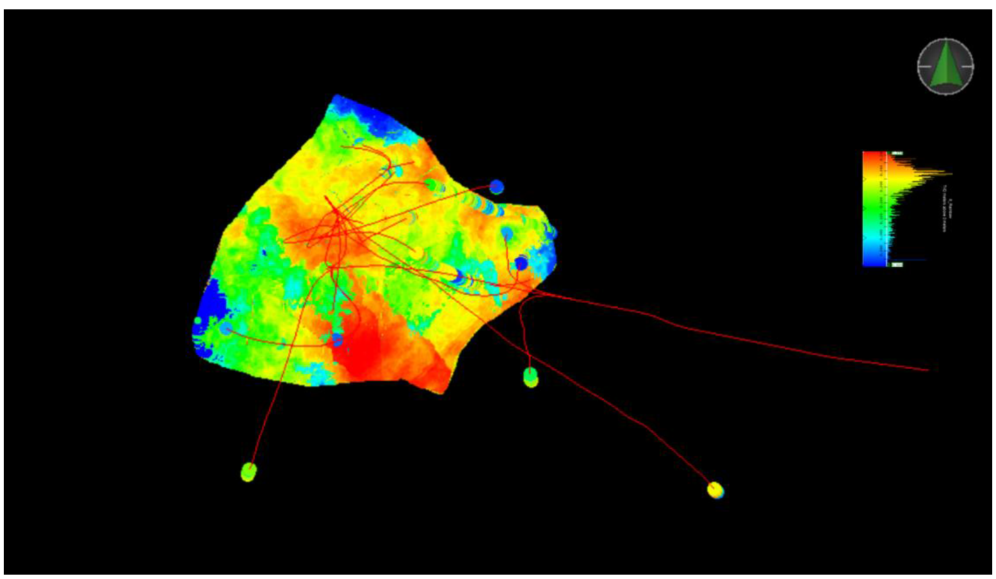

Figure 7—An example of Porosity model built with the facies constraint in the background

Novel/ Additive Workflow

One of the novel workflows used here is Scalable Earth Modeling which is a grid-less modeling approach which builds the consistent multi-resolution geo-model. This is followed by implementing another innovative workflow solution Unified Ensemble Modeling which creates an orchestrated workflow to run the static and dynamic uncertainty modeling together to come up with history matched static and dynamic pairs to complete the feedback loop. The value propositions or differentiators which have been attained through the "Ensemble based Modeling guided by Grid-less static model approach are as below:

1. Incorporating multi-purpose model into a single "Shared Earth Container" model without compromising the resolution issue
2. Optimizing the resolution and model extent with respect to the hardware capability without losing information of the original input data through upscaling
3. Generating multiple static models with different resolution but with geological consistency
4. Improved Geostatistical analysis capturing the input data in its original form
5. Model update with various investigative scenarios such as introducing the shale layer towards the northern part of the model or capturing thin layers as per the Reservoir Engineer's advice or introducing new fault barriers were seamless and rapid.
6. Integrated uncertainty analysis combining the domains like Geophysics, Petrophysics, Geological modeling and reservoir Engineering to capture the entire spectrum of uncertainty in the final results
7. An Automated process of Data Driven History Matching workflow
8. Analytics Based decision making combining all the 80 Ensemble models instead of the decision process based on rejection criteria

Conclusion and Way Forward

In summary, combining the Unified Ensemble Modeling approach with Scalable Earth Modeling provides a platform where all data that informs the subsurface understanding can be integrated in a consistent workflow, so that models are a reasonable representation using data from all subsurface disciplines and across all scales. This approach allows subsurface teams to embrace uncertainties in the reservoir modeling and history matching process.

The novel workflow that has been implemented in this project can further be enriched by integrating the Full-field well planning followed by the production forecast of the designed well and field development scenario tied to the field economics. This approach can complement the whole workflow and create a full Field Development Alternatives evaluation in a much faster and seamless way.

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